

2021 JC2 CT3 Solution

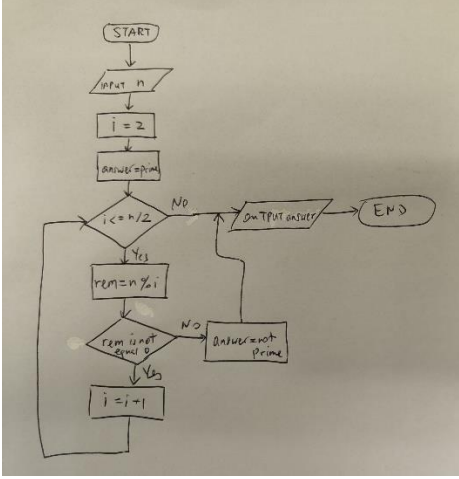
1

[6]

Conditions	C1 GRR G	C2 GRRB Impossible case	C3 GRG R	C4 GRB R	C5 WRR W	C6 WRRB Impossible case	C7 WRW R	C8 WRB R
Gold	Y	Y	Y	Y	N	N	N	N
At least one consecutive draws of the same colour	Y	Y	N	N	Y	Y	N	N
Fewer than 3 colours Runs out of ball	Y	N	Y	N	Y	N	Y	N
Outcomes								
Grand	X	X	X	X	X			
\$20	X	X			X	X		
\$50	X		X		X		X	

Conditions	C1/5	C2 GRRB	C3 GRGR	C4 GRBR		C6 WRRB	C7 WRWR	C8 WRBR
Gold	-	Y	Y	Y		N	N	N
At least one consecutive draws of the same colour	Y	Y	N	N		Y	N	N
Fewer than 3 colours Runs out of ball	Y	N	Y	N		N	Y	N
Outcomes								
Grand	X	X	X	X				
\$20	X	X				X		
\$50	X		X				X	

No	Solution	Marks
2a	[9, 4, 6, 2, 7, 1, 8, 5]	1
2b	list index out of range	1
2c/d	<pre>def delete(lst, item): if len(lst) == 0: return [] if lst[0] == item: return delete(lst[1:], item) else: return [lst[0],] + delete(lst[1:], item)</pre>	1 1
2e		5

	 <pre> graph TD START([START]) --> Input[/Input n/] Input --> i2[i = 2] i2 --> LoopCond{i <= n/2} LoopCond -- No --> OutputPrime[/Output prime/] OutputPrime --> END([END]) LoopCond -- Yes --> RemCalc[rem = n % i] RemCalc --> RemCheck{rem is not equal to 0} RemCheck -- Yes --> OutputNotPrime[/Output not prime/] RemCheck -- No --> iInc[i = i + 1] iInc --> LoopCond </pre>	
3a-e	a) As the network administrator of a boarding school, you are tasked to restrict boarders from using the school LAN resources to access online gambling website (e.g. https://www.singaporepools.com.sg/) and the use of insecure file transfer protocol (e.g. FTP). Explain how you can achieve this. [2] • Configure firewall to block outgoing web requests based on ip address • Configure firewall to block port number of insecure file transfer protocol (e.g. 20 and 21) b) What is the purpose of a network switch? [2] • A network switch connects devices within a network to form a local area network, or LAN • forwards data packets to and from those devices in the same LAN. c) State the most important protocol in the Internet layer, explain how it achieves its goal. [3] • Internet protocol • relaying datagrams/packets across network boundaries • IP has the task of delivering packets from the source host to the destination host solely based on the IP addresses in the packet headers d) What is circuit switching? [1] • a method of implementing a telecommunications network in which two network nodes establish a dedicated communications channel (circuit) through the network before the nodes may communicate. e) Explain how encryption helps in network security. [2] • protect the confidentiality of digital data either stored on computer systems (General hacking of information in database) • protect the confidentiality of digital data transmitted through a network like the Internet.	

	(Man in the middle attack)	
4a	1 mark for 3 classes 1 mark for correct use of public and private 1 mark for correct distribution of attributes 1 mark for identification of appropriate methods 1 mark for correct inheritance shown (upward pointing arrows) 1 mark for polymorphism (circle display()) <pre> class Staff - name: str - contact: str - bank_acct: str + Staff (name: str, contact: str, bank_acct: str) + set_contact (new_contact: str) + get_contact (): str + display(): str class Contract_Staff - hourly_rate: float - records: list = [] + Contract_Staff (name: str, contact: str, bank_acct: str, hourly_rate: float) + add_new_record (new_record: tuple) + get_records(): list + display(): str class Perm_Staff - name: str - department: str - monthly_rate: float - bonus: float = 0.0 + Perm_Staff (name: str, contact: str, bank_acct: str, monthly_rate: float) + set_bonus (new_bonus: float) + get_bonus(): float + display(): str </pre>	[6]
b	Bundling of data and methods, provide data hiding. Example Public method to control user behavior, such as only provide get method for bank account with certain restrictions.	[3]
c	Sub class will auto inherit all attributes and methods from superclass. Example Promote code reuse.	[3]
d	05 & 06 Syntax error is a structural error of a program such that the code violates grammar/rules of the programming language. Change RECORD(0)to RECORD[0] 08 Runtime error refers to error which was not detected when the program was compiled but is only revealed when a particular line is executed. 03 Change to DECLARE TOTAL_DURATION: FLOAT = 0 11 Logic error is one which allows a program to run successfully but produces an unintended or undesired result. 11 Change to RETURN TOTAL_DURATION * HOURLY_RATE	[6]
e	Try Except block/Set breakpoint to capture run time error at 06 Print statement before line 11	[4]

f	Tuple is a rigid implementation, if there's any errors, it's hard to make corrections. Only date and duration are stored means limited information is captured. Create a class containing all necessary information and related operations	[3]																
5a	<pre> 2 / \ 1 6 / \ 4 9 / \ 3 5</pre>	2																
b	<pre> 4 / \ 2 6 / \ 1 9 / \ 3 5</pre>	2																
c	<pre>1 2 3 4 5 6 9</pre>	1																
d	1 3 5 4 9 6 2	1																
e	BST allows retrieval of its contents in order in O(N) time, while Hash table contents needs to be sorted first before it can be retrieved in order.	1																
f	No need goes for optimal/best time complexity. If student mentions retrieval the BST data in-order O(N), follow by a rebuilding of the BST with insertion of the middle items of the ordered list repeated recursively will do. It will take O(NlogN) time <- inserting N item and each insertion takes O(lgN). no penalty for wrong time complexity	3																
6a	Hash function key % 7 <table><thead><tr><th>slot</th><th>key</th></tr></thead><tbody><tr><td>0</td><td>76</td></tr><tr><td>1</td><td></td></tr><tr><td>2</td><td></td></tr><tr><td>3</td><td>10</td></tr><tr><td>4</td><td>31</td></tr><tr><td>5</td><td></td></tr><tr><td>6</td><td>6</td></tr></tbody></table>	slot	key	0	76	1		2		3	10	4	31	5		6	6	4
slot	key																	
0	76																	
1																		
2																		
3	10																	
4	31																	
5																		
6	6																	
b	No account of the position of the letters is made. Anagrams have the same hash value. EVIL and LIVE.	1																
c	<pre>>>> lst = [6,2,3,5,4] >>> bubbleSort(lst)</pre> <pre>[6, 3, 2, 5, 4] [6, 3, 5, 2, 4] [6, 3, 5, 4, 2] [6, 5, 3, 4, 2] [6, 5, 4, 3, 2]</pre> For those sorted in wrong direction -1 <pre>[2, 6, 3, 5, 4]</pre>	3																
		2																

	<pre> [2, 3, 6, 5, 4] [2, 3, 5, 6, 4] [2, 3, 5, 4, 6] [2, 3, 4, 5, 6] </pre>	
d	<pre> def onepass(lst, i): if i > len(lst) - 2: return False else: if lst[i] > lst[i+1]: return onepass(lst, i+1) else: # perform swap on the i th and i+1 th item lst[i], lst[i+1] = lst[i+1], lst[i] return True or onepass(lst, i+1) </pre>	1 2 3
e	<pre> 1,16 1,7 1,3 3,3 </pre>	3
f	• There is no code to handle item not present, recursion continues indefinitely • Each recursive call results in use of storage for current state of program • Eventually memory will be exhausted / heap exhausted / stack overflow	3
7a	UNF BookingID, BookingDate, StartDate, NumberOfLessons, CustomerID, CustomerName, CustomerContact, CustomerEmail, TeacherID, TeacherName, TeacherContact, TeacherEmail, LessonID1, Content1, Instrument1, LessonDate1, LessonTime1, LessonVenue1, LessonID2, Content2, Instrument2, LessonDate2, LessonTime2, LessonVenue2... 1NF: BookingID, LessonID, BookingDate, StartDate, NumberOfLessons, CustomerID, CustomerName, CustomerContact, CustomerEmail, TeacherID, TeacherName, TeacherContact, TeacherEmail, Content, Instrument, LessonDate, LessonTime, LessonVenue 2NF Booking (BookingID, BookingDate, StartDate, NumberOfLessons, CustomerID, CustomerName, CustomerContact, CustomerEmail, TeacherID, TeacherName, TeacherContact, TeacherEmail) BookingLesson (BookingID*, LessonID*, LessonDate, LessonTime, LessonVenue) Lesson (LessonID, Content, Instrument) 3NF: Customer (CustomerID, CustomerName, CustomerContact,	[6]

	CustomerEmail) Teacher (TeacherID, TeacherName, TeacherContact, TeacherEmail) Booking (BookingID, BookingDate, StartDate, NumberOfLessons, CustomerID*, TeacherID*) BookingLesson (BookingID*, LessonID*, LessonDate, LessonTime, LessonVenue) Lesson (LessonID, Content, Instrument) Legend: Underline – Primary Key Star* - Foreign Key	
b	Customer 1:n Booking n:1 BookingLesson n:1 Lesson n :1 Teacher	[3]
c	Unique, one or more, not null Example Primary key with more than one field Example	[4]
d	Data Integrity refers to a requirement for data to be accurate and up to date. Flat files have data redundancy, and hence hard to maintain all data to be accurate and up to date when inserting/updating/deleting data. DBMS stores data in its normalized form, with minimal data redundancy, this feature makes it easier to maintain data integrity.	[3]
e	Customer (CustomerID, CustomerName, CustomerContact) Teacher (TeacherID, TeacherName, TeacherContact) Booking (BookingID, CustomerID*, TeacherID*) BookingLesson (BookingID*, LessonID*, LessonDate, LessonTime, LessonVenue)	[4]